DIY ESP32 MARAUDER MINI

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ESP32 Marauder Mini

Wi-Fi & BLE Security Analysis Device

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Disclaimer

This project documentation is created strictly for educational and demonstration purposes as part of a cybersecurity learning. The ESP32 Marauder Mini is an open-source security research tool, and this build represents a personal hardware implementation and documentation of the project. All tests, demonstrations, and configurations were performed in controlled environments and on networks or devices owned by the author or with proper authorization. The techniques and tools described in this report are intended only for ethical cybersecurity research, awareness, and defensive learning. Any misuse of the information or tools presented in this document for unauthorized access, disruption, or illegal activity is strictly discouraged, and the author takes no responsibility for such actions.

Open Source Attribution

This project is based on the **ESP32 Marauder** open-source firmware developed by **"JustCallMeKoko."** The firmware provides a portable platform for wireless security testing and research.

The following points clarify the nature of this project:

- The original firmware and concept belong to the open-source project by **JustCallMeKoko**.
- This work focuses on **hardware assembly, configuration, and practical implementation** of the Marauder Mini version.
- No claim is made over the original software, codebase, or design.
- This documentation is created solely for **educational, and demonstration purposes**.

All credit for the firmware, core features, and original project design goes to the original developer and contributors.

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INTRODUCTION

The ESP32 Marauder Mini is a compact, portable cybersecurity device designed for wireless network analysis, testing, and educational purposes. It is based on the ESP32 microcontroller and runs open-source firmware developed by the cybersecurity community. The device is capable of performing various Wi-Fi and Bluetooth scanning operations, making it a useful tool for learning about wireless security concepts in a practical and hands-on manner.

With the growing importance of wireless communication in everyday life, understanding how wireless networks function and how they can be secured has become essential for aspiring cybersecurity professionals. The ESP32 Marauder project provides an affordable and accessible platform for experimenting with network reconnaissance techniques, packet monitoring, and device interaction in a controlled environment.

This project focuses on building a **DIY version of the ESP32 Marauder Mini**, starting from a breadboard prototype and later converting it into a compact, functional hardware model. The objective was to gain practical experience in hardware assembly, firmware configuration, troubleshooting, and documentation of a real-world cybersecurity tool.

Through this project, the developer explored various aspects of embedded systems, wireless networking, and open-source cybersecurity tools. The process involved selecting suitable components, wiring the circuit, installing the firmware, testing the device's functionality, and documenting the entire development cycle.

The ESP32 Marauder Mini project not only demonstrates technical skills in electronics and firmware handling but also reflects an understanding of ethical cybersecurity practices. All testing performed using this device is intended strictly for educational and authorized environments.

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PROJECT OVERVIEW

The ESP32 Marauder Mini is a handheld wireless network testing device built using an ESP32 development board and a small TFT display. It runs open-source firmware that provides a menu-driven interface for performing different wireless scanning and analysis tasks.

This project involved creating a DIY prototype of the Marauder Mini using commonly available electronic components. The initial design was developed on a breadboard to test connections, firmware compatibility, and basic functionality. After successful testing, the device was assembled into a more stable and organized working model.

The system integrates multiple hardware modules, including the ESP32 microcontroller, a TFT display for the user interface, 5 way joystick switch module, a GPS module, and supporting components such as a battery, step-up module, and SD card interface. These components work together to provide a portable and interactive platform for wireless network analysis.

The firmware used in this project is based on the open-source ESP32 Marauder project. It provides features such as Wi-Fi scanning, Bluetooth device detection, packet monitoring, and other testing utilities through a graphical menu interface.

The main focus of this project was not to create new firmware, but to understand, build, configure, and document a complete working device from scratch. This included hardware assembly, firmware flashing, troubleshooting errors, optimizing connections, and producing structured technical documentation.

The final outcome of this project is a functional DIY ESP32 Marauder Mini device along with a detailed report that explains the design process, configuration steps, testing methods, and lessons learned during development.

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EXECUTIVE SUMMARY

The DIY ESP32 Marauder Mini project is a compact, portable wireless security assessment device built using an ESP32 microcontroller and open-source Marauder firmware. The primary goal of this project was to understand the practical implementation of wireless penetration testing concepts by assembling, configuring, and testing a handheld device capable of performing multiple wireless analysis tasks.

The device was designed using an ESP32 development board, a 1.44-inch TFT display, push buttons for navigation, and additional modules such as a GPS receiver and SD card interface. The system was first developed on a breadboard to test circuit connections and firmware compatibility. After successful testing, the prototype was transferred to a more stable hardware setup to create a clean and portable working model.

The firmware used in this project is the open-source ESP32 Marauder firmware developed by JustCallMeKoko. This firmware provides a wide range of Wi-Fi and Bluetooth Low Energy (BLE) scanning, monitoring, and testing features. The device can perform operations such as Wi-Fi network scanning, packet monitoring, BLE device discovery, GPS-based wardriving and related attacks. These capabilities make it a useful educational tool for understanding wireless communication protocols and security concepts.

The project emphasizes hands-on learning in areas such as embedded systems, firmware flashing, hardware interfacing, debugging, wireless security testing and attacks. Through this project, the developer gained practical experience in assembling electronic components, configuring open-source firmware, and performing controlled wireless testing in a lab environment.

Overall, the DIY ESP32 Marauder Mini serves as a compact and cost-effective platform for experimenting with wireless security tools while demonstrating practical skills in both hardware and cybersecurity domains.

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TECHNOLOGY OVERVIEW

The ESP32 Marauder Mini is built around the ESP32 microcontroller, which is a low-cost, low-power system-on-chip with integrated Wi-Fi and Bluetooth capabilities. This microcontroller acts as the core processing unit of the device, handling wireless scanning, packet analysis, user interface control, and interaction with external modules.

The device uses a TFT display to provide a graphical interface for navigating menus, viewing scan results, and controlling different features. 5 way joystick switch module is connected to the ESP32's GPIO pins to allow user input without requiring a touchscreen. This switch-based interface makes the device simple, durable, and suitable for compact portable builds.

A GPS module is integrated into the system to enable wardriving functionality. Wardriving involves collecting information about wireless networks along with geographic coordinates. The GPS module communicates with the ESP32 using serial communication, allowing the device to log network data with location information.

An SD card module is used for data storage. This allows the device to save captured wireless data, scan results, or wardriving logs for later analysis. The SD card communicates with the ESP32 using the SPI interface.

The firmware powering the device is the open-source ESP32 Marauder firmware. This firmware provides multiple wireless analysis features, including Wi-Fi scanning, packet monitoring, BLE scanning, and other security testing tools.

The development process involved tools such as the ESP Web Tool for flashing, USB-to-serial communication for programming the ESP32, and basic electronic prototyping tools such as breadboards, jumper wires, and soldering equipment.

Overall, the technology stack combines embedded hardware, open-source firmware, wireless communication protocols, and development tools to create a compact wireless security testing platform.

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SYSTEM IMPLEMENTATION

Hardware Design and Integration

The ESP32 Marauder Mini was implemented as a compact embedded security device by integrating multiple hardware components around the ESP32 development board. The design process began with a breadboard-based configuration to validate electrical connections, communication protocols, and peripheral compatibility before moving to a permanent build.

The TFT display was interfaced using the SPI protocol, with dedicated GPIO lines for MOSI, SCLK, CS, DC, RST, and backlight control. An SD card module was also connected via SPI for data logging, with a separate chip-select line to prevent bus conflicts. The NEO-6M GPS module was integrated through UART communication to enable real-time location tracking for wardriving operations.

5 way switch module connected to designated GPIO pins to allow menu navigation and device control. Proper resistor configurations were implemented to ensure stable digital inputs and eliminate floating states. Power was supplied using a 3.7V lithium battery connected to a step-up converter to provide regulated 5V output, along with a physical on/off switch for device control.

After successful validation on the breadboard, the circuit was transferred to a double-sided vero board to create a more compact and reliable working model. Components were arranged strategically to reduce wiring clutter and improve mechanical stability. The final working model demonstrated stable booting, consistent display rendering, and reliable peripheral communication, forming a functional handheld network analysis device.

The table below presents the finalized GPIO mapping and wiring configuration used in the ESP32 Marauder Mini implementation.

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TFT-Display

Display Pins	Esp 32
VCC	3V3
GND	GND
SCL / SCK	GPIO 18
SDA / MOSI	GPIO 23
RES / RST	GPIO 05
DC / RS	GPIO 16
CS	GPIO 17
BLK / LED	GPIO 18

SD Card Module

SD Pins	Esp 32
VCC	3V3
GND	GND
CS	GPIO 12
MOSI	GPIO 23
SCK	GPIO 18
MISO	GPIO 19

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GPS Module

GPS Pins	Esp 32
VCC	3V3
GND	GND
TX	GPIO 21
RX	GPIO 22

5 Way Switch Module

Switch Pins	Esp 32
+ (VCC)	3V3
- (GND)	GND
UP	GPIO 36 (VP)
DOWN	GPIO 35
LEFT	GPIO 13
RIGHT	GPIO 39 (VN)
CENTER / OK	GPIO 34

Based on the finalized GPIO mapping, a circuit diagram was prepared to visually represent the system architecture and signal flow between components.

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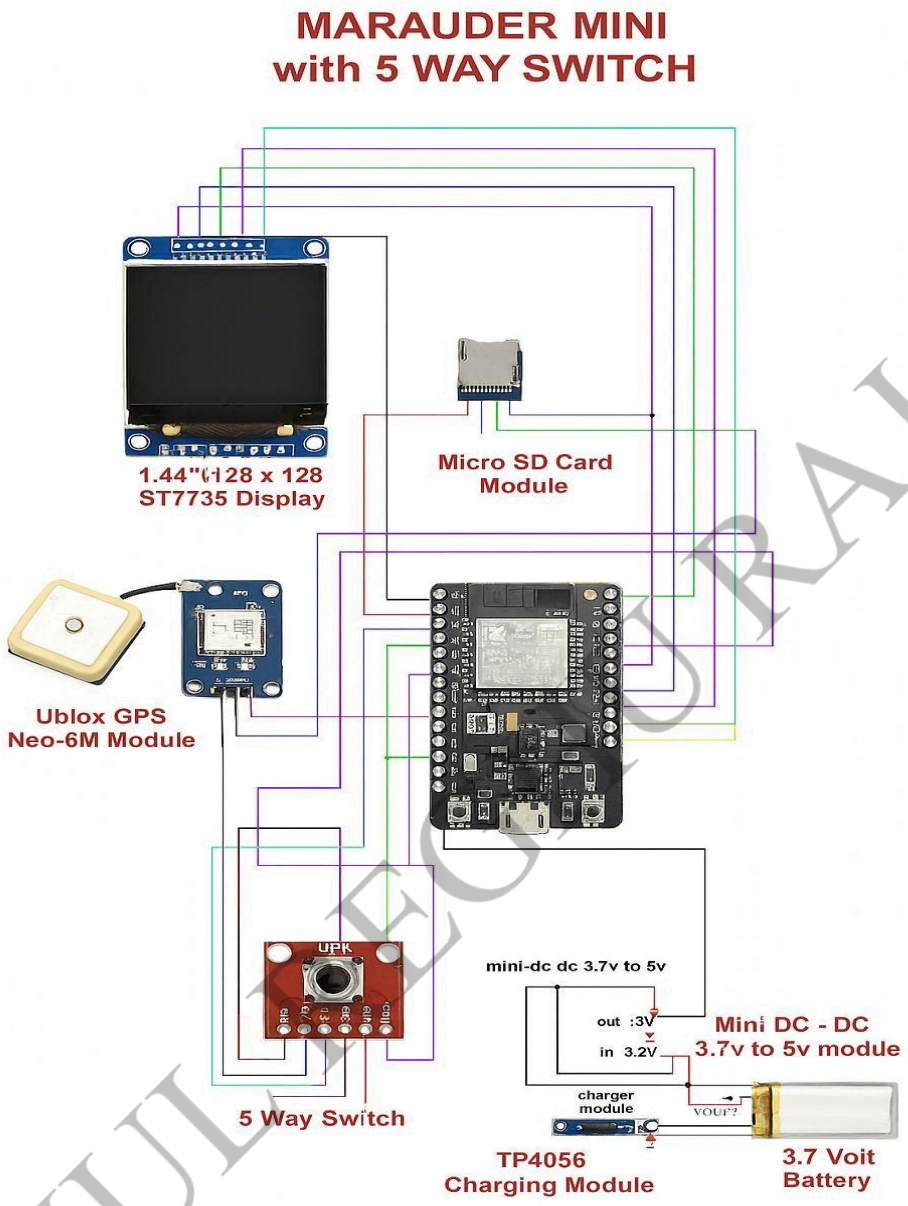

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Firmware Installation and Deployment

The ESP32 Marauder Mini operates using the official precompiled "Mini" firmware release available from the project's GitHub repository maintained by JustCallMeKoko. No source code modifications or custom firmware changes were performed for this implementation. Instead, the stable Mini firmware package was used to ensure compatibility with the selected hardware configuration.

The firmware was flashed onto the ESP32 Dev kit using the ESP Web Tool via USB-to-serial communication. The flashing process involved loading the required binary files provided in the release package, including the bootloader, partition table, boot application, and the main Mini firmware binary. Each file was assigned to its respective flash memory address as specified in the official documentation (for example, bootloader at 0x1000, partition table at 0x8000, boot app at 0xE000, and the main firmware at 0x10000).

After successful flashing, the ESP32 booted directly into the Marauder Mini interface the firmware automatically initialized the display, navigation controls, Wi-Fi, BLE, and other supported modules.

Features and Functional Capabilities

Upon successful deployment, the device supports multiple wireless security analysis functions, including:

- Wi-Fi network scanning and access point enumeration
- BLE device scanning and advertisement analysis
- Packet monitoring and capture logging
- GPS-enabled wardriving with location tagging
- SD card data storage for captured logs
- Menu-based interface navigation using hardware buttons

The system is capable of performing both passive monitoring and active wireless testing operations, making it a portable Wi-Fi and BLE security analysis tool. The integration of GPS enhances its capability for geographical mapping of wireless networks.

The completed system demonstrates a combination of embedded hardware design, firmware configuration, debugging, and cybersecurity-oriented implementation.

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TESTING METHODOLOGY

The testing phase of the ESP32 Marauder Mini project was conducted with a dual focus:

1. Hardware and firmware stability validation, and
2. Controlled wireless security testing in an ethical environment.

All wireless testing activities were performed on authorized networks and test devices to ensure compliance with ethical cybersecurity practices.

Hardware and Input Stability Testing

Initial testing revealed instability in the navigation input system, including random or “ghost” button presses. Through systematic debugging, it was identified that GPIO pins 34, 35, 36, and 39 are input-only pins and do not support internal pull-up resistors.

Without proper pull-up configuration, these pins remained in a floating state, resulting in undefined logic levels.

To resolve this:

- GPIO behavior was analyzed using serial debugging.
- Button inputs were isolated and tested individually.
- Pull-up capable GPIOs were reassigned where required.
- Proper resistor logic was implemented to ensure stable HIGH/LOW states.

This testing phase reinforced the importance of defined logic states and correct GPIO selection in embedded systems.

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Firmware and System Stability Testing

Once hardware stability was ensured, firmware behavior was validated by:

- Repeated boot cycle testing
- Menu navigation responsiveness checks
- Display initialization consistency
- GPS signal acquisition and coordinate validation
- SD card logging verification

Multiple runtime sessions were conducted to confirm that no unexpected crashes or peripheral conflicts occurred during prolonged operation.

Wireless Security Feature Testing

In a controlled lab setup using authorized Wi-Fi networks and devices, the following cybersecurity-related features were tested:

- Wi-Fi access point enumeration
- Packet monitoring and network traffic observation
- BLE device discovery and advertisement scanning
- Deauthentication testing within a permitted test network
- Beacon frame transmission analysis
- GPS-based wardriving functionality for mapping detected networks

The objective of this testing phase was to evaluate how wireless attack simulation tools operate at the packet level and analyze how client devices respond to management frame manipulation.

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This phase strengthened practical understanding of:

- Wireless attack vectors
- Network reconnaissance techniques
- BLE advertisement structures
- Defensive implications of deauthentication and spoofing attacks

The testing methodology ensured that the project was validated both as a reliable embedded system and as a controlled wireless security testing platform. The structured debugging and ethical attack simulations provided hands-on experience in both hardware troubleshooting and cybersecurity concepts.

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CHALLENGES FACED AND LEARNING OUTCOMES

The development of the ESP32 Marauder Mini involved several technical challenges during hardware integration, firmware behavior analysis, and system stabilization. These challenges required structured troubleshooting and careful review of ESP32 architecture and firmware logic.

Challenges Faced

During the development process, the following challenges were encountered:

1. GPIO Pull-Up Errors on Input-Only Pins
GPIO pins 34, 35, 36, and 39 were initially used for navigation inputs. These pins are input-only and do not support internal pull-up resistors. This resulted in unstable input behavior.

2. Floating Input Pins Causing Ghost Button Presses
Due to lack of proper pull-up configuration, certain GPIOs remained in a floating state. This caused random or unintended button triggers during operation.

3. Confusion Between Pull-Up Capable and Non-Pull-Up GPIOs
Understanding the ESP32 pin architecture required deeper review of documentation to distinguish between fully functional GPIOs and restricted input-only pins.

4. Black Screen During Boot Initialization
During startup, the TFT display flashed white and then turned black, initially appearing as a firmware failure. After systematic observation and testing, it was discovered that the firmware required the center (OK) button to be pressed during boot for proper initialization.

5. Ensuring Stable Multi-Module Integration
Integrating the TFT display, GPS module, SD card module, and navigation controls required careful wiring validation and repeated boot cycle testing to ensure stable communication.

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Learning Outcomes

From the challenges encountered during the development and testing of the ESP32 Marauder device, the following cybersecurity-focused learning outcomes were achieved:

• Developed practical understanding of how embedded hardware platforms such as the ESP32 can be utilized to perform wireless security testing and network reconnaissance.

• Gained hands-on experience with Wi-Fi packet monitoring, wireless environment scanning, and the role of management frames in wireless network communication.

• Improved understanding of how penetration testing tools operate at the hardware level, including the interaction between firmware, wireless chipsets, and supporting modules such as GPS and SD logging.

• Learned the importance of stable hardware configuration for reliable execution of wireless security testing operations such as packet sniffing, network enumeration, and signal monitoring.

• Acquired practical knowledge of debugging hardware issues that could affect wireless testing accuracy, including GPIO configuration, power stability, and boot behavior analysis.

• Strengthened analytical thinking required for cybersecurity research by adopting a structured troubleshooting methodology rather than relying on trial-and-error debugging.

• Developed deeper insight into how embedded systems can support real-world cybersecurity experimentation and security research environments.

Overall, this project reinforced the relationship between embedded hardware engineering and practical cybersecurity testing, demonstrating how reliable hardware design directly supports effective wireless security analysis.

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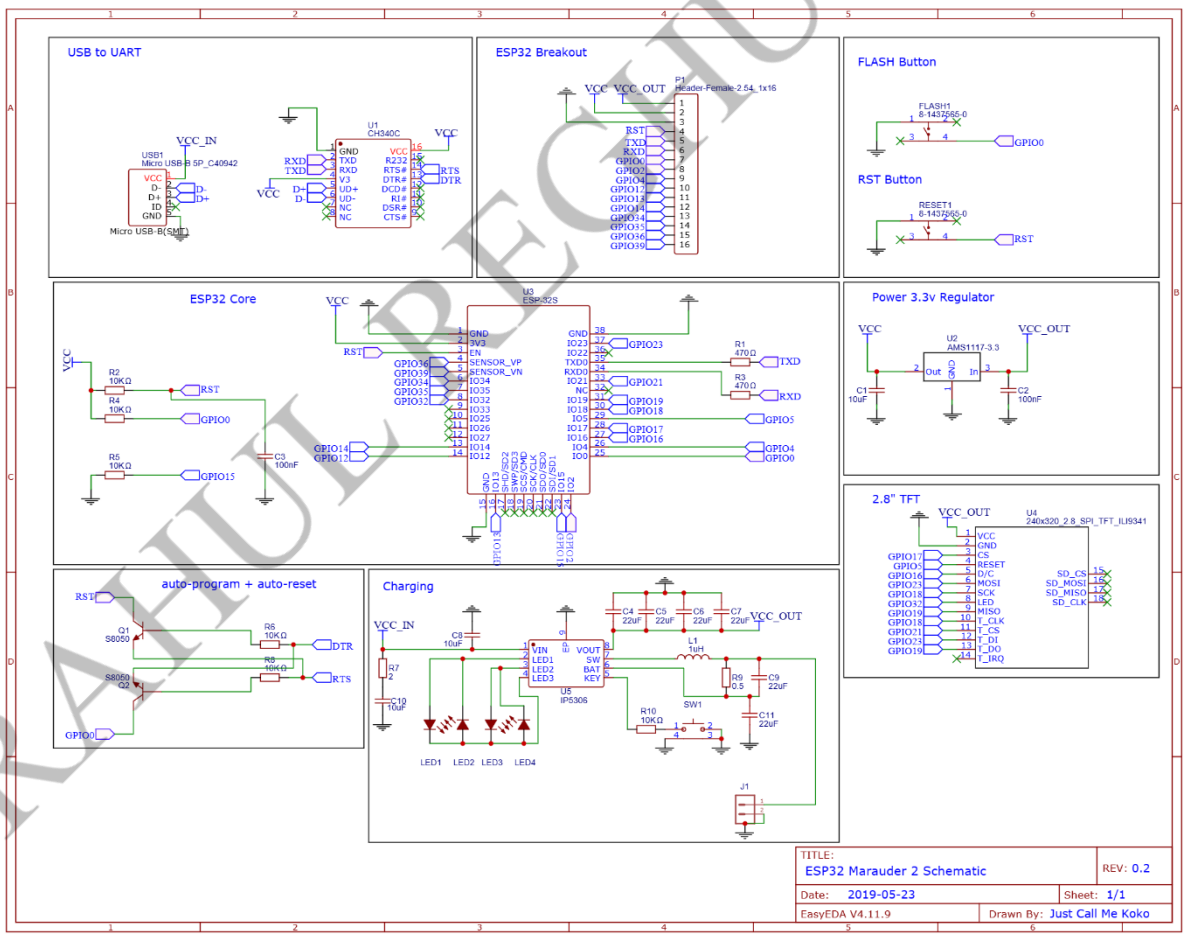
APPENDIX A - PROTOTYPE IMAGES

This appendix provides visual documentation of the hardware development process, including the breadboard proof-of-concept stage and the finalized working model. These images serve as supporting evidence of the practical implementation and physical assembly of the ESP32 Marauder Mini.

Reference Schematic Diagram

ESP32 Marauder Reference Schematic (Original Design by JustCallMeKoko)



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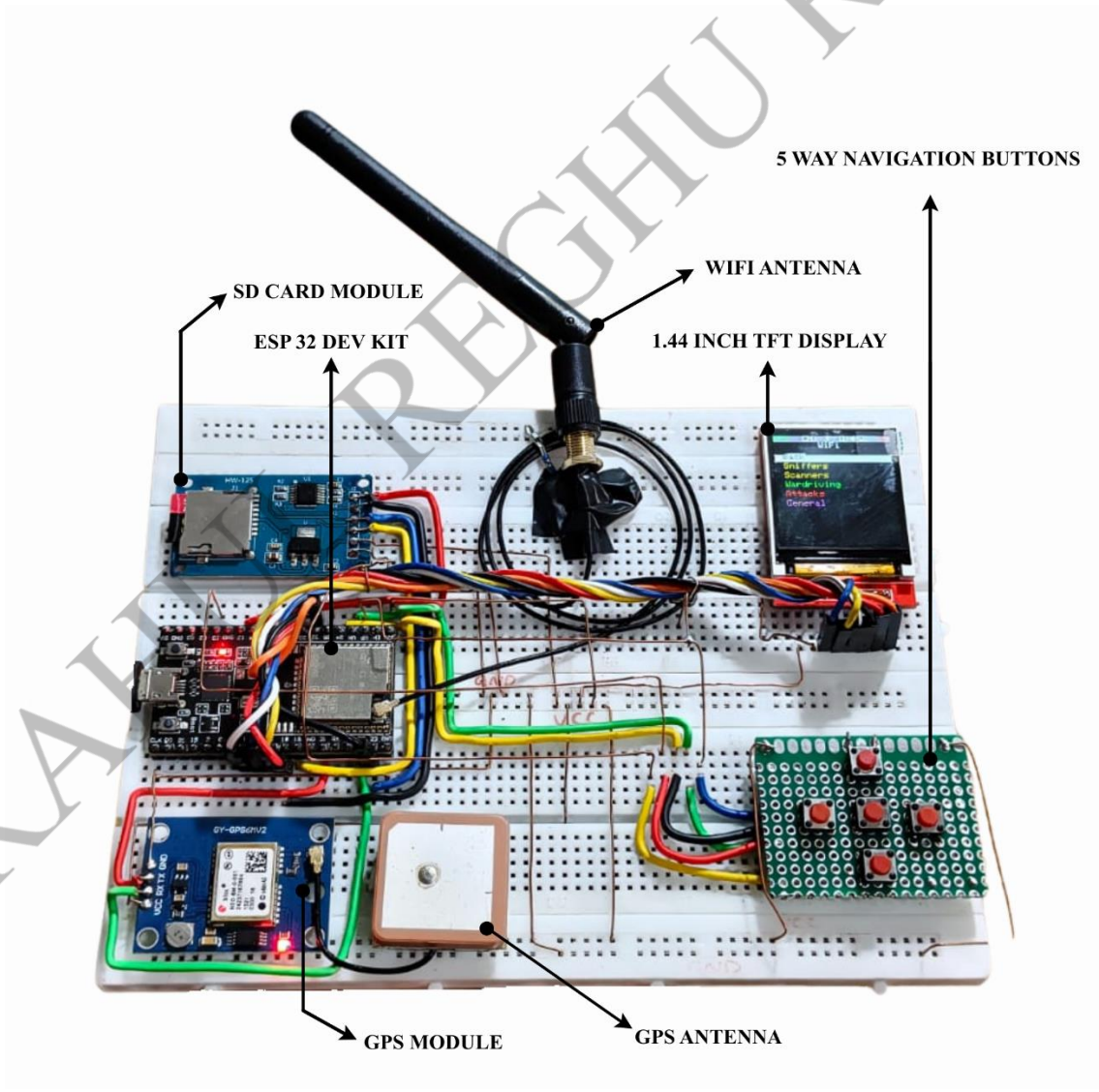
The above schematic represents the original ESP32 Marauder hardware design created by JustCallMeKoko. It illustrates the ESP32 core connections, USB-to-UART interface, power regulation circuit, auto-reset circuitry, charging module, and TFT display integration.

This schematic was used as a technical reference during the hardware implementation phase of the DIY ESP32 Marauder Mini build to understand GPIO mapping, power distribution, and peripheral interfacing principles.

Breadboard Prototype (Proof of Concept)



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Initial breadboard-based prototype used for hardware validation and GPIO testing. The prototype stage was used to verify input stability, SPI communication, and firmware boot behavior before final assembly.

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Final Working Model

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Final assembled ESP32 Marauder Mini with integrated display, GPS module, navigation switch, and portable power system.

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APPENDIX B – ENGAGEMENT INFORMATION

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CONCLUSION

The DIY ESP32 Marauder Mini project successfully combined embedded systems implementation with practical wireless security testing concepts. Through the integration of the ESP32 Dev Kit, TFT display, GPS module, SD card logging, navigation controls, and portable power management, a fully functional handheld Wi-Fi and BLE security analysis device was developed.

The project required careful hardware integration, understanding of ESP32 GPIO architecture, structured troubleshooting, and controlled validation of wireless testing features. Challenges related to floating GPIO inputs, boot behavior, and multi-module stability provided valuable hands-on learning in embedded debugging and system reliability.

Beyond hardware implementation, the project strengthened practical understanding of wireless reconnaissance, packet monitoring, BLE advertisement analysis, and ethical attack simulation within authorized environments. It demonstrated how embedded platforms can be leveraged for cybersecurity research and controlled testing scenarios.

Overall, this project reflects the integration of engineering discipline, structured problem-solving, and cybersecurity awareness. It represents a practical step forward in developing both embedded systems expertise and wireless security knowledge.

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